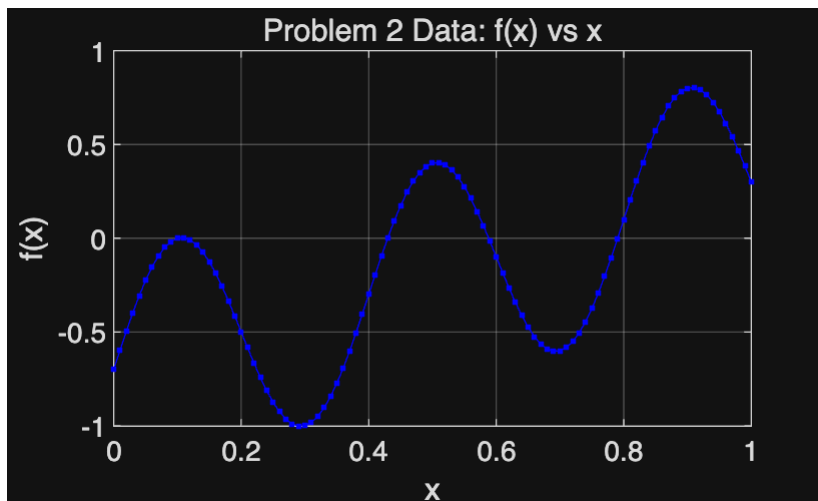


HW3 Question 2

```
clear; clc; close all;

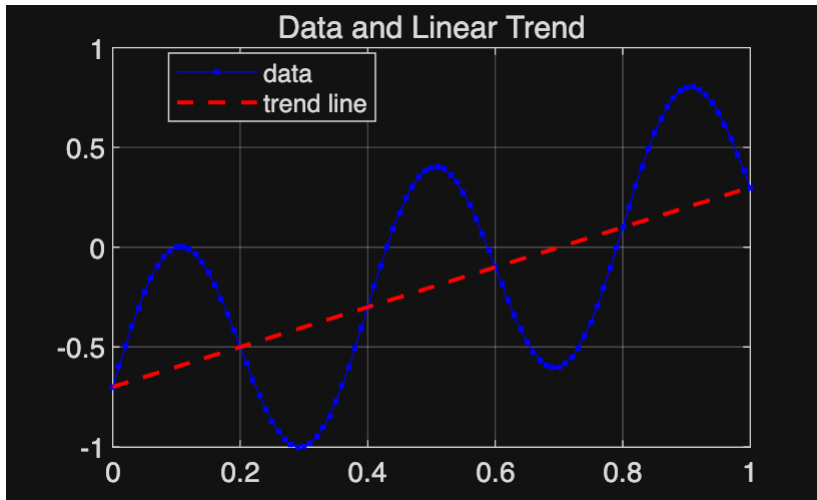
x = 0:0.01:1;
y = load("problem2.txt");
y = y(:);
N = numel(y);
assert(numel(x) == N, 'x and y must have the same length (101).');

figure(1);
plot(x, y, 'b.-');
xlabel('x'); ylabel('f(x)');
title('Problem 2 Data: f(x) vs x');
grid on;
```

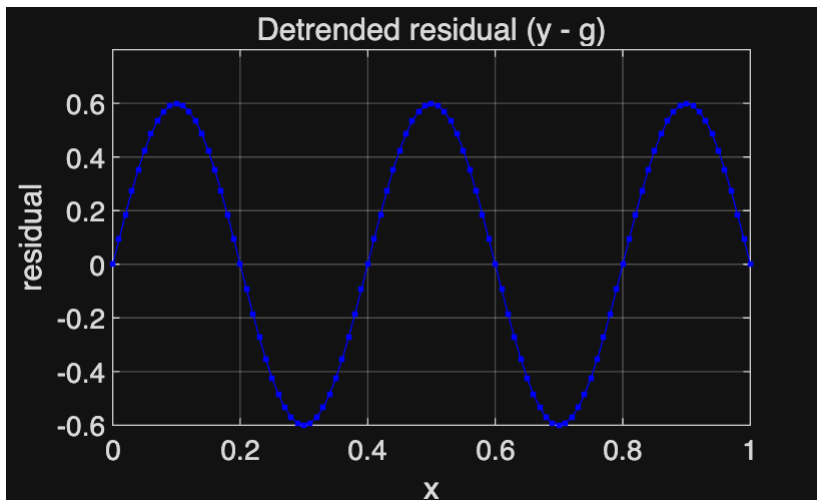


```
y0 = y(1);
y1 = y(end);
g = (y1 - y0) * x + y0;

figure(2);
plot(x, y, 'b.-'); hold on;
plot(x, g, 'r--', 'LineWidth', 1.5);
legend('data', 'trend line', 'Location', 'best');
title('Data and Linear Trend');
grid on;
```



```
t = y - g.';
figure(3);
plot(x, t, 'b.-');
xlabel('x'); ylabel('residual');
title('Detrended residual (y - g)');
grid on;
```



Now we need to approximate the curve. Since the curve has 2.5 period in the range, we could pick $k = 2.5$

```
k = 2.5;
Ak = [cos(2*pi*k*x(:)), sin(2*pi*k*x(:))];
beta = (Ak.'*Ak) \ (Ak.'*t);
bc = beta(1);
bs = beta(2);

yhat = g.' + Ak*beta;
```

```

a0 = y(1);
a1 = y(end) - y(1);
fprintf('k = %.3f\n', k);

```

```

k = 2.500

```

```

fprintf('a0 = %11.4e\n', a0);

```

```

a0 = -7.0000e-01

```

```

fprintf('a1 = %11.4e\n', a1);

```

```

a1 = 1.0000e+00

```

```

fprintf('bc = %11.4e\n', bc);

```

```

bc = -5.4530e-17

```

```

fprintf('bs = %11.4e\n', bs);

```

```

bs = 6.0000e-01

```

```

res = y - yhat;
mse = mean(res.^2);
R2 = 1 - sum(res.^2)/sum((y-mean(y)).^2);
fprintf('MSE = %.4e, R^2 = %.6f\n', mse, R2);

```

```

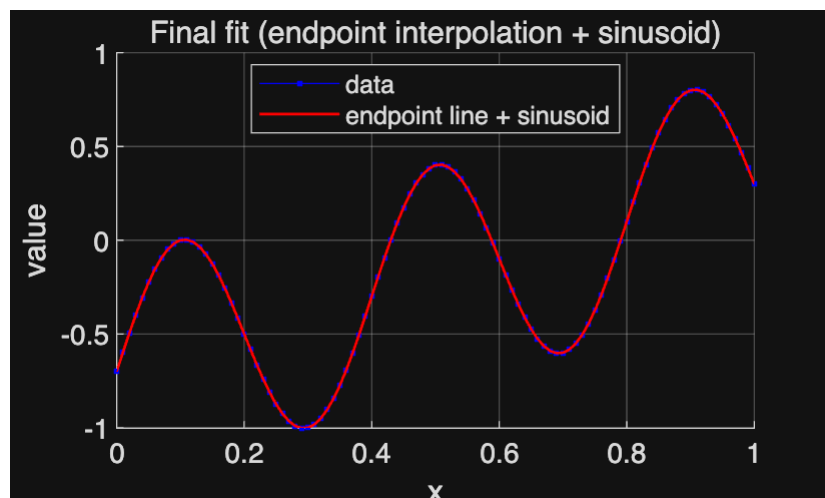
MSE = 3.3357e-31, R^2 = 1.000000

```

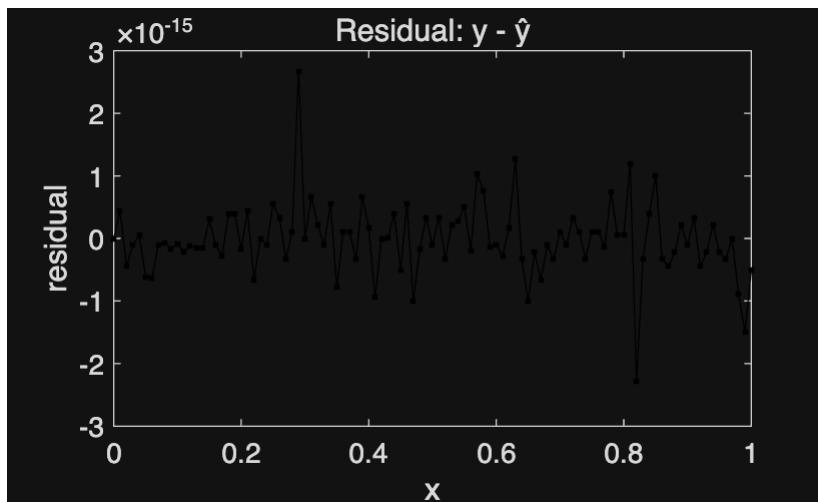
```

figure; hold on; grid on;
plot(x, y, 'b.-', 'DisplayName', 'data');
plot(x, yhat, 'r-', 'LineWidth', 1.0, 'DisplayName', 'endpoint line + sinusoid');
legend('Location', 'best');
xlabel('x'); ylabel('value');
title('Final fit (endpoint interpolation + sinusoid)');

```



```
figure; grid on;
plot(x, res, 'k.-');
xlabel('x'); ylabel('residual');
title('Residual:  $y - \hat{y}$ ');
```



```
fprintf('\nFinal fitted function:\n');
```

Final fitted function:

```
fprintf('f(x) = %.4e + (%.4e)*x + (%.4e)*cos(%.4f*pi*x) + (%.4e)*sin(%.4f*pi*x)\n',
...
      a0, a1, bc, 2*k, bs, 2*k);
```

```
f(x) = -7.0000e-01 + (1.0000e+00)*x + (-5.4530e-17)*cos(5.0000*pi*x) + (6.0000e-01)*sin(5.0000*pi*x)
```

Convert to amplitude–phase

```
w = 2*pi*k;
A = hypot(bc, bs);
phi = atan2(bs, bc);
fprintf('Equivalent: f(x) = %.4e + (%.4e)*x + %.4e*cos(%.4f*x - %.4f)\n', ...
      a0, a1, A, w, phi);
```

```
Equivalent: f(x) = -7.0000e-01 + (1.0000e+00)*x + 6.0000e-01*cos(15.7080*x - 1.5708)
```

Error Metrics

```
res = y - yhat;
mse = mean(res.^2);
rmse = sqrt(mse);
R2 = 1 - sum(res.^2)/sum((y-mean(y)).^2);
fprintf('RMSE = %.4e, MSE = %.4e, R^2 = %.6f\n', rmse, mse, R2);
```

```
RMSE = 5.7755e-16, MSE = 3.3357e-31, R^2 = 1.000000
```

